

Proof of Theorem on Sampling Distribution

Theorem 1. Let $X_1, X_2, \dots, X_n$ be i.i.d. $N(\mu, \sigma^2)$ random variables. Let $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$ and $S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$. Then $\bar{X}$ and S^2 are independently distributed and

$$\bar{X} \sim N(\mu, \sigma^2/n) \quad \text{and} \quad \frac{(n-1)S^2}{\sigma^2} \sim \chi_{n-1}^2.$$

Proof. Let A be an $n \times n$ orthogonal matrix, whose first row is

$$\left(\frac{1}{\sqrt{n}}, \frac{1}{\sqrt{n}}, \dots, \frac{1}{\sqrt{n}} \right).$$

Note that such a matrix exists as we can start with the row and construct a basis of $\mathbb{R}^n$. Then Gram-Schmidt orthogonalization will give us the required matrix. As A is orthogonal, its inverse exists and $A^{-1} = A^T$, the transpose of A . Now consider the transformation of random vector $\mathbf{X} = (X_1, X_2, \dots, X_n)'$ given by

$$\mathbf{Y} = A\mathbf{X}.$$

First, we shall try to find the distribution of $\mathbf{Y}$. Note that the transformation $g(\mathbf{x}) = A\mathbf{x}$ is a one-to-one transformation as A is invertible. The inverse transformation is given by $\mathbf{x} = A'\mathbf{y}$. Hence, the Jacobian of the inverse transformation is $J = \det(A)$. As A is orthogonal, absolute value of $\det(A)$ is one. Now, as $X_1, X_2, \dots, X_n$ are i.i.d. $N(\mu, \sigma^2)$ RVs, the JPDP of $\mathbf{X}$, for $\mathbf{x} = (x_1, x_2, \dots, x_n)' \in \mathbb{R}^n$, is

$$\begin{aligned} f_{\mathbf{X}}(\mathbf{x}) &= \frac{1}{(\sigma\sqrt{2\pi})^n} \exp \left[-\frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2 \right] \\ &= \frac{1}{(\sigma\sqrt{2\pi})^n} \exp \left[-\frac{1}{2\sigma^2} (\mathbf{x} - \boldsymbol{\mu})'(\mathbf{x} - \boldsymbol{\mu}) \right], \end{aligned}$$

where $\boldsymbol{\mu} = (\mu, \mu, \dots, \mu)'$ is a n component vector. Thus, the JPDP of $\mathbf{Y}$, for $\mathbf{y} \in \mathbb{R}^n$, is

$$\begin{aligned} f_{\mathbf{Y}}(\mathbf{y}) &= f_{\mathbf{X}}(A'\mathbf{y}) \\ &= \frac{1}{(\sigma\sqrt{2\pi})^n} \exp \left[-\frac{1}{2\sigma^2} (A'\mathbf{y} - \boldsymbol{\mu})'(A'\mathbf{y} - \boldsymbol{\mu}) \right] \\ &= \frac{1}{(\sigma\sqrt{2\pi})^n} \exp \left[-\frac{1}{2\sigma^2} (\mathbf{y} - \boldsymbol{\eta})'(\mathbf{y} - \boldsymbol{\eta}) \right], \end{aligned}$$

where $\boldsymbol{\eta} = (\eta_1, \eta_2, \dots, \eta_n)' = A\boldsymbol{\mu}$. Note that $\eta_1 = \sqrt{n}\mu$. Moreover,

$$\boldsymbol{\eta}'\boldsymbol{\eta} = \boldsymbol{\mu}'\boldsymbol{\mu} \implies \sum_{i=1}^n \eta_i^2 = n\mu^2 \implies \sum_{i=2}^n \eta_i^2 = n\mu^2 - \eta_1^2 = 0.$$

Thus, $\eta_i = 0$ for $i = 2, 3, \dots, n$. Hence the JPDP of $\mathbf{Y}$ is

$$f_{\mathbf{Y}}(\mathbf{y}) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2\sigma^2}(y_1 - \sqrt{n}\mu)^2} \left\{ \prod_{i=2}^n \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{y_i^2}{2\sigma^2}} \right\} \quad \text{for } \mathbf{y} = (y_1, y_2, \dots, y_n)' \in \mathbb{R}^n.$$

Therefore, $Y_1, Y_2, \dots, Y_n$ are independent RVs and $Y_1 \sim N(\sqrt{n}\mu, \sigma^2)$ and $Y_i \sim N(0, \sigma^2)$ for $i = 2, 3, \dots, n$, where $\mathbf{Y} = (Y_1, Y_2, \dots, Y_n)'$. Now,

$$Y_1 = \sqrt{n}\bar{X} \implies \sqrt{n}\bar{X} \sim N(\sqrt{n}\mu, \sigma^2) \implies \bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right).$$

Again,

$$\mathbf{Y}'\mathbf{Y} = \mathbf{X}'\mathbf{X} \implies \sum_{i=2}^n Y_i^2 = \sum_{i=1}^n X_i^2 - Y_1^2 = \sum_{i=1}^n X_i^2 - n\bar{X}^2 = (n-1)S^2.$$

For $i = 2, 3, \dots, n$, $\frac{Y_i}{\sigma}$ are i.i.d. $N(0, 1)$ RVs. Thus, using the previous theorem

$$\frac{(n-1)S^2}{\sigma^2} = \sum_{i=2}^n \left(\frac{Y_i}{\sigma}\right)^2 \sim \chi_{n-1}^2.$$

Notice that $\bar{X}$ is a function of Y_1 only, and S^2 is a function of $Y_2, Y_3, \dots, Y_n$. As Y_i 's are independent, $\bar{X}$ and S^2 are independent. □